# A flat elliptical function in $\mathbb{R}^4$ that is not a finite sum of regular squares

Candidate solution packet for arXiv:2107.12840

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**Abstract**

Korobenko and Sawyer call a function *regular* when it belongs to $C^{2,\delta}$ for some $\delta > 0$, and construct elliptical flat smooth functions not representable as finite sums of regular squares in dimensions at least five. We give such a function already on $\mathbb{R}^4$. The extra scale coordinate in their construction is replaced by a sequence of disjoint hard four-dimensional islands accumulating at the origin. Their compactness lower bound is diagonalized simultaneously over the number of squares and a countable cofinal family of Hölder exponents. No monotonicity property is claimed.


**Status and scope.** This is a candidate counterexample. It closes the unconstrained dimension-four existence gap left by the source's $n \geq 5$ theorem. It does *not* settle the separate question in source Remark 2.7 asking whether specified weak monotonicity hypotheses force a positive result in dimensions $2, 3, 4$.

## 1 Source input

Set

$$L(w, x, y, z) = w^4 + x^2y^2 + y^2z^2 + z^2x^2 - 2wxyz. \tag{1}$$

For $0 < \alpha < 1$, let $\mathcal{C}_{\nu,\alpha}(\tau)$ denote the lower-bound function supplied by Korobenko–Sawyer, Lemma 5.2, for the modulus $\omega(t) = t^\alpha$. The only source fact we use is

$$\mathcal{C}_{\nu,\alpha}(\tau) \longrightarrow \infty \quad (\tau \downarrow 0), \tag{2}$$

and every identity on the unit ball

$$L + \tau = \sum_{j=1}^{\nu} h_j^2 \tag{3}$$

with $h_j \in C^{2,\alpha}$ satisfies

$$\sum_{j=1}^{\nu} \|h_j\|_{C^{2,\alpha}(B(0,1))} \geq \mathcal{C}_{\nu,\alpha}(\tau). \tag{4}$$

This is a qualitative compactness assertion; no rate in $\tau$ is needed. Notice also that $L \geq 0$: AM–GM applied to the four nonnegative monomials in (1) gives a sum at least $4|wxyz|$, which remains nonnegative after subtracting $2wxyz$.

## 2 Proof intuition

One hard copy of $L$ multiplied by a flat factor does not suffice: after normalizing the copy, derivatives of hypothetical square roots may acquire an arbitrarily large inverse flat factor. We instead put a separate copy $L+\tau_n$ in each of infinitely many tiny disjoint balls. At the $n$-th ball the normalization loss is a finite number $R_n$, however large. We then choose $\tau_n$ so small that the source lower bound exceeds $n^2R_n$ for every number of squares and every Hölder exponent among the first $n$ possibilities. If one finite regular-square decomposition existed, its rescaled norms would be at most $CR_n$ on all late islands, contradicting the chosen lower bound.

The island amplitudes are exponentially smaller than every power of their radii. Consequently all cutoff errors, including derivatives of every order, disappear to infinite order at the sole accumulation point.

## 3 The counterexample

**Theorem 1.** *There exists $f \in C^\infty(\mathbb{R}^4;[0,\infty))$ such that*

1. *$f(0)=0$, $f(x)>0$ for every $x\neq 0$, and every derivative of $f$ vanishes at the origin;*
2. *on no neighbourhood of the origin can $f$ be written as $f=\sum_{j=1}^{\nu} g_j^2$ for finitely many functions $g_j$, each of which belongs to $C^{2,\delta_j}$ for some $\delta_j>0$.*

*In particular, $f$ is an elliptical flat smooth function on $\mathbb{R}^4$ that is not a finite sum of regular squares.*

*Proof.* Write $e_1=(1,0,0,0)$, and for $n\geq 1$ put

$$r_n = 2^{-n-4}, \qquad p_n = r_n e_1, \qquad \rho_n = \frac{r_n^2}{100}, \qquad a_n = \exp(-\rho_n^{-2}). \tag{5}$$

Choose $\chi \in C_c^\infty(\mathbb{R}^4)$ with $0\leq\chi\leq 1$, equal to one on $B(0,1)$, and supported in $B(0,2)$. Set $\chi_n(x)=\chi((x-p_n)/\rho_n)$. The closed supports are pairwise disjoint: consecutive centers have distance $r_n/2$, whereas

$$2\rho_n + 2\rho_{n+1} = \frac{r_n^2}{40} < \frac{r_n}{2}.$$

They also avoid the origin and accumulate only there.

For $k,\nu\in\mathbb{N}$, abbreviate $\mathcal{C}_{\nu,k}=\mathcal{C}_{\nu,1/k}$, taking $k\geq 2$ when a Hölder exponent below one is required. Define the normalization loss

$$R_n = 1 + \frac{\rho_n r_n + \rho_n^2}{\sqrt{a_n}}. \tag{6}$$

By (2), for every $n$ we may choose $0<\tau_n<1/n$ so small that

$$\mathcal{C}_{\nu,k}(\tau_n) \geq n^2 R_n \quad \text{for all } 1\leq\nu\leq n, \quad 2\leq k\leq n. \tag{7}$$

Only finitely many conditions occur at each stage.

Let

$$b(x) = \begin{cases} \exp(-|x|^{-2}), & x\neq 0, \\ 0, & x=0, \end{cases} \qquad F_n(x) = a_n\left[L\left(\frac{x-p_n}{\rho_n}\right)+\tau_n\right],$$

and define

$$f(x) = b(x) + \sum_{n=1}^{\infty} \chi_n(x)\big(F_n(x)-b(x)\big). \tag{8}$$

Because the supports are disjoint, at most one summand is nonzero at each point. On a support, $f=(1-\chi_n)b+\chi_nF_n$. Both terms being positive away from zero (and $F_n>0$ because $\tau_n>0$ and $L\geq 0$), equation (8) is nonnegative and is strictly positive off the origin.

We next justify smoothness at the accumulation point. On $B(p_n, 2\rho_n)$ one has $|x| \asymp r_n$. For every derivative order $m$ and every $N$, flatness of $b$ gives

$$\sup_{B(p_n,2\rho_n)} |D^m b| = O_{m,N}(r_n^N).$$

Leibniz' rule, $D^q\chi_n = O_q(\rho_n^{-q})$, and the fixed degree of $L$ give, for every $m$,

$$\sup_{B(p_n,2\rho_n)} |D^m[\chi_n(F_n - b)]| \le C_m a_n \rho_n^{-m} + O_{m,N}(r_n^N \rho_n^{-m}). \tag{9}$$

Since $\rho_n = r_n^2/100$ and $a_n = \exp(-10^4 r_n^{-4})$, the right side is $o(r_n^A)$ for every prescribed $A$, after choosing $N$ large in the second term. Thus every derivative of the correction in (8) extends continuously to zero with value zero. Standard induction on the derivative order shows that $f \in C^\infty$ and all its derivatives vanish at zero.

It remains to exclude regular squares. Suppose on some ball about zero that

$$f = \sum_{j=1}^{\nu} g_j^2, \qquad g_j \in C^{2,\delta_j}, \quad \delta_j > 0. \tag{10}$$

Let $\delta = \min_j \delta_j$, and fix an integer $k \ge 2$ with $\alpha = 1/k < \delta$. On a smaller closed ball, all $g_j$ have bounded $C^{2,\alpha}$ norms; denote a common bound by $K$.

From (10), $|g_j| \le \sqrt{f}$. Since $f$ is flat, $\sqrt{f} = o(|x|^M)$ for every $M$. Hence every $g_j$ is flat as a function, and its $C^2$ Taylor expansion implies

$$g_j(0) = 0, \qquad Dg_j(0) = 0. \tag{11}$$

For all sufficiently large $n$, the inner ball $B(p_n, \rho_n)$ lies in the decomposition domain and $n \ge \max\{\nu, k\}$. There $\chi_n = 1$, so after writing $x = p_n + \rho_n W$ and setting

$$G_{j,n}(W) = \frac{g_j(p_n + \rho_n W)}{\sqrt{a_n}}, \tag{12}$$

we obtain on $B(0,1)$

$$L(W) + \tau_n = \sum_{j=1}^{\nu} G_{j,n}(W)^2. \tag{13}$$

The function term in the $C^{2,\alpha}$ norm is uniformly bounded because $|G_{j,n}| \le \sqrt{L + \tau_n}$. From (11), the mean value theorem, and the chain rule,

$$\begin{aligned}
\|DG_{j,n}\|_\infty &\le CK \frac{\rho_n(r_n + \rho_n)}{\sqrt{a_n}}, \\
\left\|D^2 G_{j,n}\right\|_\infty &\le K \frac{\rho_n^2}{\sqrt{a_n}}, \\
[D^2 G_{j,n}]_\alpha &\le K \frac{\rho_n^{2+\alpha}}{\sqrt{a_n}} \le K \frac{\rho_n^2}{\sqrt{a_n}}.
\end{aligned}$$

Consequently a constant $C_0$, depending on the alleged decomposition but not on $n$, satisfies

$$\sum_{j=1}^{\nu} \|G_{j,n}\|_{C^{2,\alpha}(B(0,1))} \le C_0 R_n. \tag{14}$$

On the other hand, the source estimate (4), the diagonal choice (7), and (13) imply that the same sum is at least $n^2 R_n$. Hence $n^2 \le C_0$ for every sufficiently large $n$, an impossibility. This proves the theorem. $\square$

# 4 Why the fifth coordinate is unnecessary here

The source uses a fifth variable continuously to encode the scale and also engineers a prescribed modulus of monotonicity. For the unconstrained existence problem, a discrete sequence of four-dimensional spatial islands does the same diagonal job. The key logical order is: first fix the island geometry and amplitude, hence the finite loss $R_n$; only then choose the floor $\tau_n$ using the unbounded compactness modulus. Reversing that order would create a circular rate requirement.

This observation does not automatically preserve any chosen $\omega$-monotonicity. Indeed, the relation between the tiny core floor $a_n\tau_n$ and surrounding values would require quantitative control of the source hardness function, while Lemma 5.2 supplies only divergence. The dimension-four monotonicity threshold therefore remains outside the claim.

# 5 Source evidence

Figure 1 records the source definitions, Theorem 2.5 with its $n \geq 5$ hypothesis, and Remark 2.7. Figure 2 records the hardness lemma used above.

SUMS OF SQUARES I 5

**Definition 2.2.** *A nonnegative function $f : B(0,a) \to \mathbb{R}$ is* flat, *or vanishes to infinite order at the origin in $\mathbb{R}^n$, if*

$$\lim_{x\to 0} |x|^{-N} f(x) = 0, \qquad \text{for all } N \in \mathbb{N}.$$

**Definition 2.3.** *A function $g : \mathbb{R}^n \to \mathbb{R}$ is* regular *if $g \in \bigcup_{\delta>0} C^{2,\delta}(\mathbb{R}^n)$, i.e. $g$ is $C^{2,\delta}$ for some $0 < \delta < 1$.*

**Definition 2.4.** *A function $f : \mathbb{R}^n \to \mathbb{R}$ is* elliptical *if $f(x) > 0$ for $x \neq 0$, and more generally, an $N \times N$ matrix-valued function $F : \mathbb{R}^n \to \mathbb{R}^{N\times N}$ is* elliptical *if $F(x)$ is positive definite for $x \neq 0$.*

Here is a strengthening of the counterexample in [BoBrCoPe, Theorem 1.2 (d)].

**Theorem 2.5.** *Let $n \geq 5$.*

*(1) If $\beta > s > 0$, there is an elliptical, flat, smooth $\omega_s$-monotone function $f$ that cannot be written as a finite sum of squares of $C^{2,\beta}(\mathbb{R}^n)$.*

*(2) Suppose $\omega$ is a modulus of continuity such that $\omega_s \ll \omega$ for all $0 < s < 1$. Then there is an elliptical, flat, smooth $\omega$-monotone function $f$ that cannot be written as a finite sum of squares of regular functions. In particular we can take $\omega = \omega_0$.*

The following corollary highlights the sharpness of the above results within the scale of $\omega$-monotone conditions.

**Corollary 2.6.** *Suppose that $f : \mathbb{R}^n \to [0,\infty)$ is elliptical, flat and smooth.*

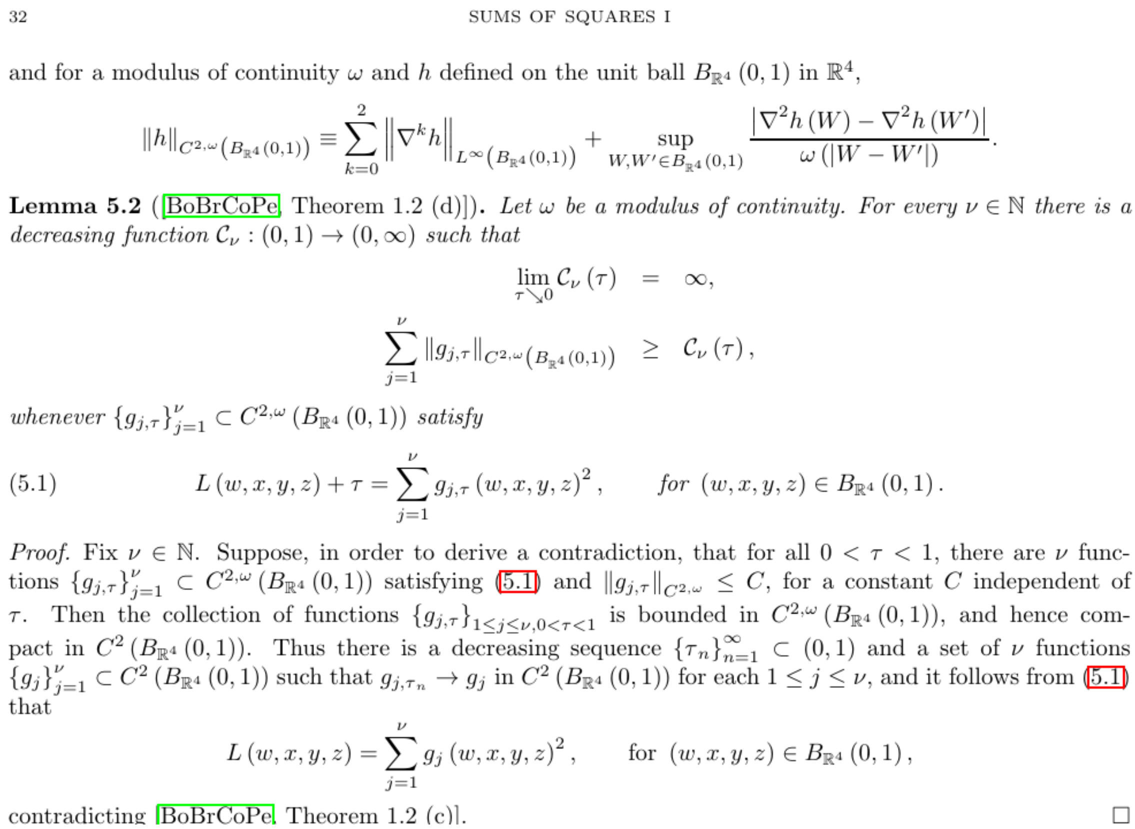
*(1) Then $f$ can written as a finite sum of squares of regular functions if $f$ is Hölder monotone.*

*(2) Conversely, for any modulus of continuity $\omega$ satisfying $\omega \gg \omega_s$ for all $0 < s < 1$, there is an $\omega$-monotone function $f$ that cannot be written as a finite sum of squares of regular functions.*

**Remark 2.7.** *The answer to Problem 2 is affirmative in all dimensions $n \geq 1$ if $f$ is Hölder monotone. However, in dimension $n = 1$, the answer is affirmative without any additional $\omega$-monotone assumptions at all [Bon], while in dimension $n \geq 5$, the assumption of Hölder monotone is essentially sharp. We do not know if significantly weaker montonicity assumptions will imply an affirmative answer to Problem 2 in the remaining dimensions $n = 2, 3, 4$.*

The following theorem on extracting smooth roots motivates our definition of nearly monotone, and is what initially led us to consider $\omega_s$-monotone functions, and which then ultimately played a key role in the above decompositions into sums of regular functions.

Figure 1: Korobenko–Sawyer, arXiv:2107.12840, PDF page 5 (cropped).



32 SUMS OF SQUARES I

and for a modulus of continuity $\omega$ and $h$ defined on the unit ball $B_{\mathbb{R}^4}(0,1)$ in $\mathbb{R}^4$,

$$\|h\|_{C^{2,\omega}(B_{\mathbb{R}^4}(0,1))} \equiv \sum_{k=0}^{2} \left\|\nabla^k h\right\|_{L^\infty(B_{\mathbb{R}^4}(0,1))} + \sup_{W,W'\in B_{\mathbb{R}^4}(0,1)} \frac{\left|\nabla^2 h(W) - \nabla^2 h(W')\right|}{\omega(|W-W'|)}.$$

**Lemma 5.2** ([BoBrCoPe, Theorem 1.2 (d)]). *Let $\omega$ be a modulus of continuity. For every $\nu \in \mathbb{N}$ there is a decreasing function $\mathcal{C}_\nu : (0,1) \to (0,\infty)$ such that*

$$\begin{aligned} \lim_{\tau \searrow 0} \mathcal{C}_\nu(\tau) &= \infty, \\ \sum_{j=1}^{\nu} \|g_{j,\tau}\|_{C^{2,\omega}(B_{\mathbb{R}^4}(0,1))} &\geq \mathcal{C}_\nu(\tau), \end{aligned}$$

*whenever $\{g_{j,\tau}\}_{j=1}^{\nu} \subset C^{2,\omega}(B_{\mathbb{R}^4}(0,1))$ satisfy*

$$L(w,x,y,z) + \tau = \sum_{j=1}^{\nu} g_{j,\tau}(w,x,y,z)^2, \qquad \textit{for } (w,x,y,z) \in B_{\mathbb{R}^4}(0,1). \tag{5.1}$$

*Proof.* Fix $\nu \in \mathbb{N}$. Suppose, in order to derive a contradiction, that for all $0 < \tau < 1$, there are $\nu$ functions $\{g_{j,\tau}\}_{j=1}^{\nu} \subset C^{2,\omega}(B_{\mathbb{R}^4}(0,1))$ satisfying (5.1) and $\|g_{j,\tau}\|_{C^{2,\omega}} \leq C$, for a constant $C$ independent of $\tau$. Then the collection of functions $\{g_{j,\tau}\}_{1\leq j\leq \nu, 0<\tau<1}$ is bounded in $C^{2,\omega}(B_{\mathbb{R}^4}(0,1))$, and hence compact in $C^2(B_{\mathbb{R}^4}(0,1))$. Thus there is a decreasing sequence $\{\tau_n\}_{n=1}^{\infty} \subset (0,1)$ and a set of $\nu$ functions $\{g_j\}_{j=1}^{\nu} \subset C^2(B_{\mathbb{R}^4}(0,1))$ such that $g_{j,\tau_n} \to g_j$ in $C^2(B_{\mathbb{R}^4}(0,1))$ for each $1 \leq j \leq \nu$, and it follows from (5.1) that

$$L(w,x,y,z) = \sum_{j=1}^{\nu} g_j(w,x,y,z)^2, \qquad \text{for } (w,x,y,z) \in B_{\mathbb{R}^4}(0,1),$$

contradicting [BoBrCoPe, Theorem 1.2 (c)]. □

Figure 2: Korobenko–Sawyer, arXiv:2107.12840, PDF page 32 (cropped): the $C^{2,\omega}$ norm and Lemma 5.2.

## References


[KS] S. Korobenko and E. T. Sawyer, *Sum of squares I: scalar functions*, arXiv:2107.12840; Journal of Functional Analysis 284 (2023), 109827. In particular Definitions 2.2–2.4, Theorem 2.5, Remark 2.7, and Lemma 5.2.

[M] S. MacDonald, *Regularity preserving sum of squares decompositions*, arXiv:2303.07998 (related polynomial obstructions; no claim of overlap with the flat isolated-zero construction above).